

ILLIA PASTUSHOK

Machine Learning Engineer & AI Tech Lead

il.pastushok@gmail.com | github.com/itilushka-red | LinkedIn | Wroclaw, Poland

PROFILE

Machine Learning Engineer and AI Team Lead with 4+ years of hands-on experience designing, building, and deploying scalable AI/ML systems from scratch to production. Proven expertise in LLMs, RAG systems, computer vision, predictive modeling, and optimization. Delivered \$800K+ in annual savings through AI-driven quality control systems and led 10+ production ML projects.

TECHNICAL EXPERTISE

Programming & Data Python, SQL, Pandas, NumPy	Machine Learning & AI PyTorch, TensorFlow, Computer Vision, NLP	LLMs & GenAI RAG, Semantic Search, Prompt Engineering, LangGraph
Cloud & MLOps AWS, Docker, Kubernetes, CI/CD	Databases PostgreSQL, MongoDB, Neo4j, Vector DBs	Backend / Fullstack FastAPI, Django, REST APIs, React

EXPERIENCE

2025 — Present	Team Lead / AI Engineer CONE RED • Wroclaw, Poland Lead design and implementation of enterprise-grade AI solutions. Architected LLM-powered systems including RAG pipelines and multi-agent workflows. Established MLOps standards for CI/CD and automated deployment.
2024 — 2025	AI Engineer CONE RED • Wroclaw, Poland Developed production ML models for computer vision, NLP, and recommendation systems. Integrated LLM APIs (GPT-5, Claude) into business workflows. Implemented containerized ML services with Docker and Kubernetes.
2023 — 2024	Quality Assurance Engineer Star Global • Wroclaw, Poland Ensured software quality and regulatory compliance for enterprise and medical-grade systems. Authored and executed 300+ test cases; supported audits with zero critical findings.
2020 — 2021	Computer Vision Software Engineer (Intern) National Center, Academy of Sciences of Ukraine • Kyiv Built OpenCV-based computer vision pipelines for scientific image analysis. Collaborated with researchers to translate scientific requirements into production-ready tools.

FEATURED PROJECTS

MANUFACTURING • AI • CV AI Quality Control Engine Enterprise CV system monitoring 50+ machines processing 10,000+ units daily. Achieved 92% defect detection accuracy. \$800K Annual Savings Python • TensorFlow • OpenCV • AWS	PHARMACEUTICAL • LLM • RAG AI Recommendation & Decision Support LLM-powered recommendation engine analyzing 10,000+ medications. Processes 500+ daily queries with <2s latency. 500+ Daily Queries GPT-5 • Pinecone • Neo4j • FastAPI • AWS
FINTECH • AUTOMATION • AI Investment Due Diligence Platform Automated company intelligence aggregation using AI. Reduced analysis time from 5 hours to 5 minutes. 60X Faster Analysis Python • LLMs • Web Scraping • PostgreSQL	HEALTHCARE • AI • CV Pediatric Myopia Treatment Platform AI-driven OCT analysis system for personalized myopia treatment in children. Processed 2,000+ scans. 94% Accuracy Python • TensorFlow • Medical Imaging
HR • AI • NLP Political Candidate Profiling Platform Multi-stage AI assessment platform for candidate evaluation. Reduced screening time by 70%. 70% Time Saved Python • GPT-4 • NLP • Django • React	B2B • GRAPH • AI Graph-Based Business Matchmaking B2B platform using graph database to connect businesses. Mapped 5,000+ businesses with 82% match rate. \$2M+ Deal Value Python • Neo4j • VoyageAI • FastAPI

EDUCATION

BSc in Medical Informatics / Biomedical Engineering

Wroclaw University of Science and Technology
2022 — 2026

- Machine Learning & Data Science
 - Medical Imaging & Computer Vision
- Statistical Modeling & Signal Processing
 - Full-Stack & Backend Engineering

CERTIFICATIONS

Python for Data Science, AI & Development
IBM
Introduction to Artificial Intelligence
IBM
AWS Cloud Practitioner
Amazon Web Services
TensorFlow Developer Certificate
Google
Neo4j Certified Professional
Neo4j

LANGUAGES

English	C1 — Advanced
Polish	Upper-Intermediate
Ukrainian / Russian	Native

ADDITIONAL

- Comfortable working in Agile/Scrum environments
- Experienced in presenting technical solutions to stakeholders
- Eligible to work in the EU